

```
import pandas as pd
import seaborn as sns
from matplotlib import pyplot as plt
```

```
data_df = pd.read_csv('ecommerce_dataset_10000.csv')
data_df.head()
```

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Next steps: [Generate code with data_df](#) [New interactive sheet](#)

```
data_df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10000 entries, 0 to 9999
Data columns (total 20 columns):
#   Column                Non-Null Count  Dtype
---  -
0   customer_id           10000 non-null  object
1   first_name            10000 non-null  object
2   last_name             10000 non-null  object
3   gender                10000 non-null  object
4   age_group             10000 non-null  object
5   signup_date           10000 non-null  object
6   country               10000 non-null  object
7   product_id            10000 non-null  object
8   product_name          10000 non-null  object
9   category              10000 non-null  object
10  quantity              10000 non-null  int64
11  unit_price            10000 non-null  int64
12  order_id              10000 non-null  object
13  order_date            10000 non-null  object
14  order_status          10000 non-null  object
15  payment_method        10000 non-null  object
16  rating                10000 non-null  int64
17  review_text           10000 non-null  object
18  review_id             10000 non-null  object
19  review_date           10000 non-null  object
dtypes: int64(3), object(17)
memory usage: 1.5+ MB
```

It is observed that there are no null values in the dataset.

```
data_df.describe()
```

	quantity	unit_price	rating	
count	10000.000000	10000.000000	10000.000000	
mean	3.000800	246.647200	2.993100	
std	1.416474	282.605391	1.410621	
min	1.000000	25.000000	1.000000	
25%	2.000000	80.000000	2.000000	
50%	3.000000	129.000000	3.000000	
75%	4.000000	229.000000	4.000000	
max	5.000000	999.000000	5.000000	

it is observed that there are no outliers in the dataset.

```
data_df.value_counts()
```

customer_id	first_name	last_name	gender	age_group	signup_date	country	product_id	prod
CUST5990	Gary	Mathews	Female	Adults	6/1/2025	France	PROD114	Yo
							PROD113	W T R
							PROD106	Ins
							PROD101	Sa Gal
...	...	...	...	...	...	...	...	
CUST1000	Zachary	Jones	Male	Adults	1/26/2022	India	PROD110	Har B
							PROD108	Fitbi
							PROD103	Levi
							PROD102	S Hea
							PROD101	Sa Gal

10000 rows × 1 columns

dtype: int64

```
data_df.corr(numeric_only=True)
```

	quantity	unit_price	rating
quantity	1.000000	0.012335	-0.001399
unit_price	0.012335	1.000000	0.000597
rating	-0.001399	0.000597	1.000000



it is observed that there is no correlation between the variables in the dataset.

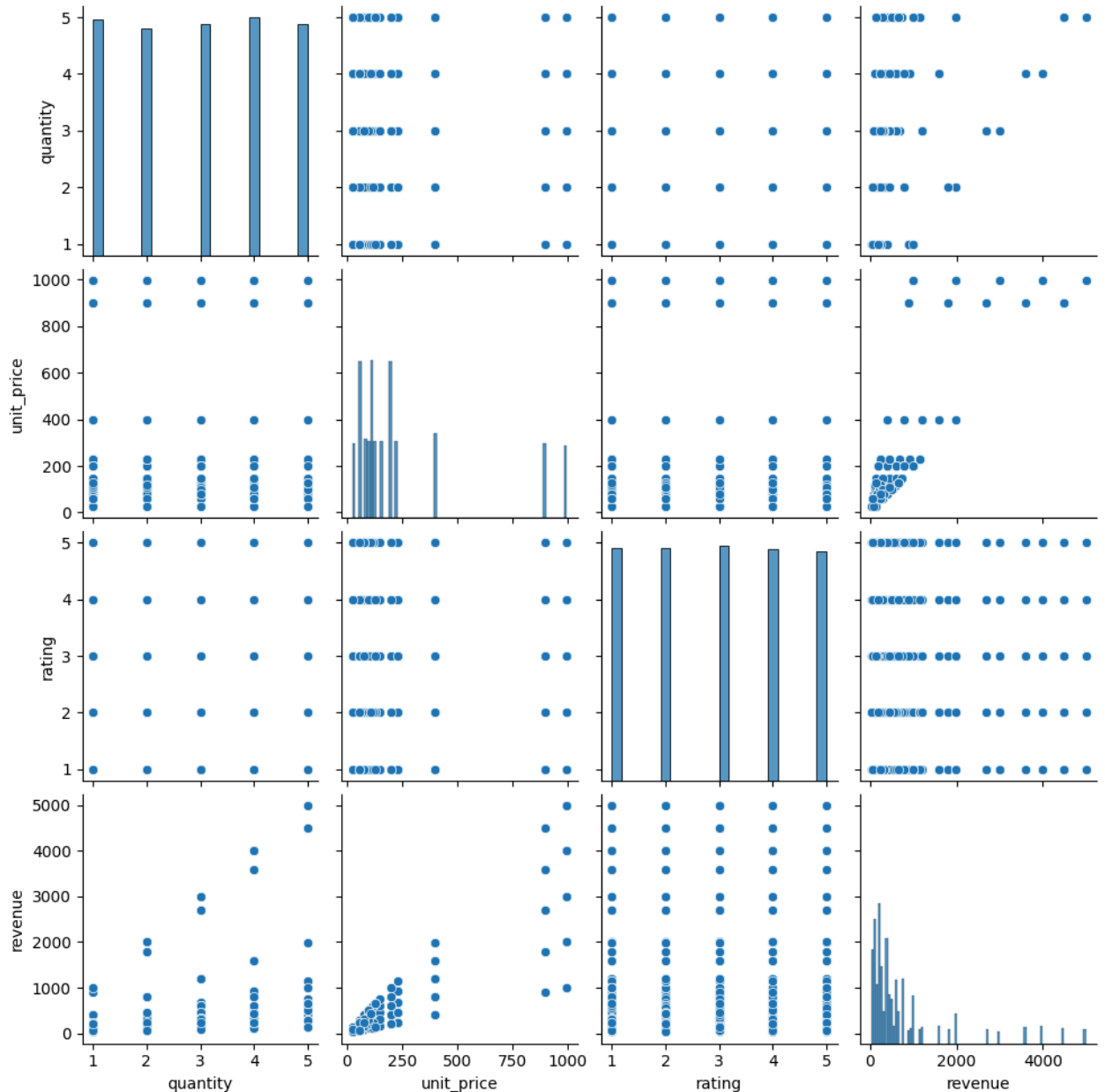
```
sns.heatmap(data_df.corr(numeric_only=True))
```

<Axes: >



```
sns.pairplot(data_df)
```

<seaborn.axisgrid.PairGrid at 0x7ee1f0157b30>



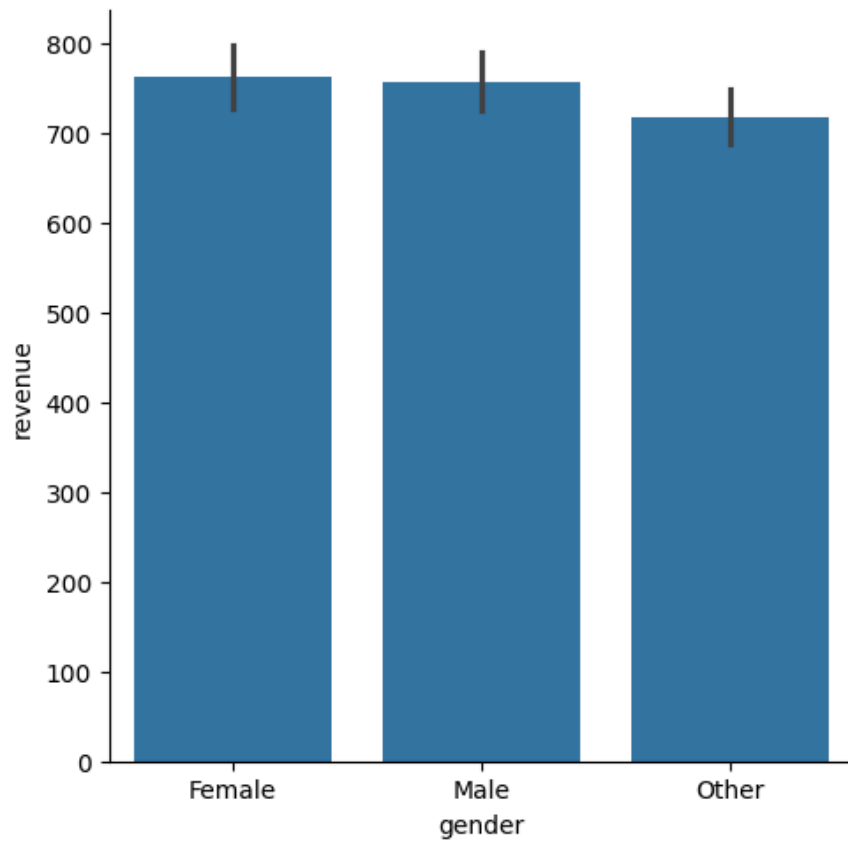
it is observed that there is no linear relationship between the variables in the dataset.

```
data_df.columns
```

```
Index(['customer_id', 'first_name', 'last_name', 'gender', 'age_group',
       'signup_date', 'country', 'product_id', 'product_name', 'category',
       'quantity', 'unit_price', 'order_id', 'order_date', 'order_status',
       'payment_method', 'rating', 'review_text', 'review_id', 'review_date',
       'revenue'],
      dtype='object')
```

```
sns.catplot(data = data_df, x = 'gender', y = 'revenue', kind = 'bar')
```

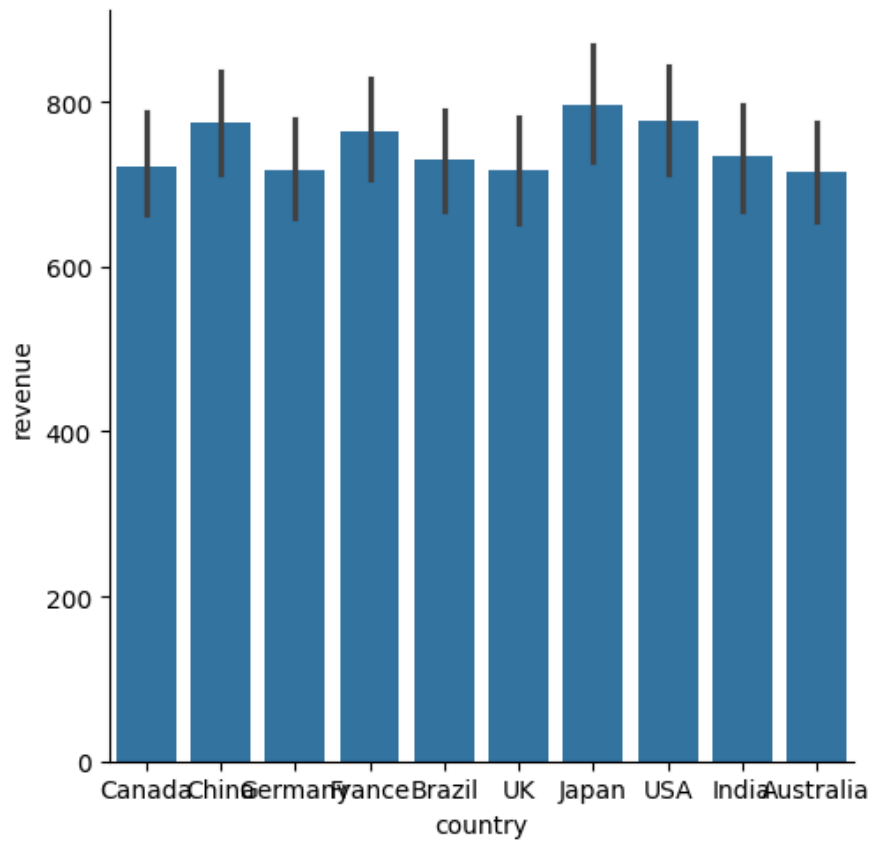
<seaborn.axisgrid.FacetGrid at 0x7ee1f15598e0>



it is observed that there is no significant difference in the revenue between the genders.

```
sns.catplot(data = data_df, x = 'country', y = 'revenue', kind = 'bar')
```

```
<seaborn.axisgrid.FacetGrid at 0x7ee1f00463c0>
```



it is observed that the revenue is highest in the Japan.

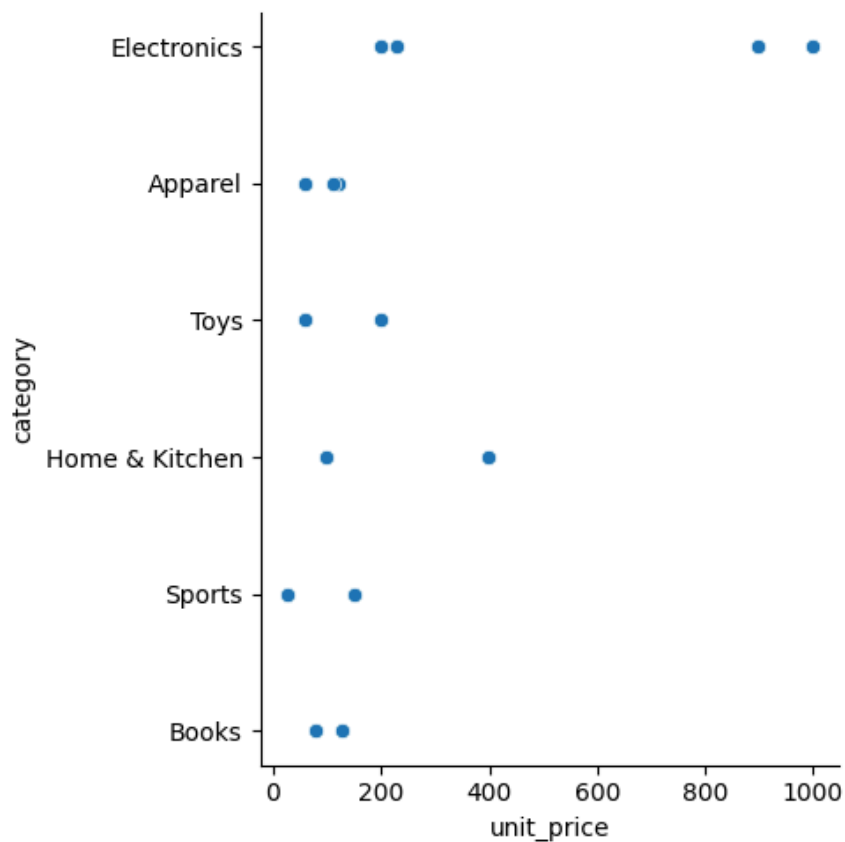
```
sns.catplot(data = data_df, x = 'age_group', kind = 'count')
```

```
<seaborn.axisgrid.FacetGrid at 0x7ee1ed50e7b0>
```

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```

```
data_df['revenue'] = data_df['quantity'] * data_df['unit_price']
sns.relplot(data = data_df, x = 'unit_price', y = 'category', kind= 'scatter')
```

```
<seaborn.axisgrid.FacetGrid at 0x7ee1f023a6f0>
```



it is observed that the revenue is highest in the category of Electronics.

```
sns.catplot(data = data_df, x = 'age_group', y = 'revenue', kind= 'box')
```

```
<seaborn.axisgrid.FacetGrid at 0x7ee1f02b4a70>
```

